

CLOUD COMPUTING & DEVOPS – AWS HANDBOOK

Student Reference Guide & Practical Journal Companion

PREFACE

Cloud Computing and DevOps have transformed the way modern applications are built, deployed, and managed. This handbook helps students understand practical objectives, maintain journals, prepare for viva examinations, and understand the complete DevOps lifecycle.

STUDENT GUIDELINES

Each student must maintain an individual journal documenting Aim, Theory, Procedure, Screenshots, Observations and Conclusion. The journal will be used during assessments, viva examinations and practical evaluations.

RESOURCE NAMING CONVENTION (MANDATORY)

All cloud resources created during practical sessions must contain the student's name. Examples: rahul-ec2, rahul-static-site, rahul-lambda, rahul-rds, rahul-app, rahul-deployment, rahul-build, rahul-pipeline.

SCREENSHOT SUBMISSION GUIDELINES

Capture screenshots showing resource names, configuration, command outputs, browser outputs, Kubernetes resources and CI/CD execution results.

JOURNAL WRITING TEMPLATE

For every practical session document: Aim, Theory, Procedure, Screenshots, Observations and Conclusion.

ACADEMIC INTEGRITY POLICY

Students must perform practical work individually. Copying screenshots or journals from classmates is prohibited.

PRACTICAL EVALUATION CRITERIA

Attendance 10%, Journal Maintenance 20%, Practical Performance 30%, Viva Examination 20%, Final Practical Test 20%.

DEVOPS LEARNING ROADMAP

Cloud Fundamentals → EC2 → S3 → Lambda → RDS → Elastic Beanstalk → Docker → Docker Networking & Storage → Docker Compose → Kubernetes → Advanced Kubernetes → Monitoring → CI/CD → End-to-End DevOps Project

SESSION 1 – CLOUD COMPUTING FUNDAMENTALS & LINUX ESSENTIALS

Why Are We Learning This?

Students must understand where applications run, how cloud infrastructure works, and why Linux powers most cloud servers.

Learning Objectives

Explain cloud computing concepts, differentiate IaaS/PaaS/SaaS, understand Linux basics, navigate the Linux filesystem and access AWS services.

Key Concepts Covered

Cloud Computing, IaaS, PaaS, SaaS, Linux Operating System, AWS Fundamentals.

Practical Activities Performed

AWS Console exploration, Linux terminal familiarization, directory management, and file operations.

Expected Outputs

AWS Services Dashboard, directory listings, file creation, file viewing and filesystem navigation outputs.

Screenshots Required

AWS Console homepage, terminal showing current directory, directory listings, file operations, and final directory structure.

Journal Observation Points

Cloud computing removes the need for physical infrastructure. Linux is widely used in cloud environments and provides command-line management capabilities.

Journal Conclusion Sample

Cloud computing provides scalable resources while Linux serves as the foundation for most cloud servers.

Viva Questions

What is cloud computing? What is IaaS? What is PaaS? What is SaaS? Why is Linux important in cloud computing?

Industry Relevance

Linux and cloud platforms are used daily by cloud engineers, DevOps engineers, SREs and cloud architects.